COMP233 — Discrete Mathematics

Lecture 4.4

Quotient-Remainder Theorem

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Number Theory and Methods of Proof

From exact division to remainders

Exact division

If d divides n, then n = dk for some integer k.
Example: 36 = 9(4).

Not exact?

An integer may leave a remainder.
Example: 38 = 9(4) + 2.

How can we describe ALL integers in this way?

The Quotient-Remainder Theorem

For every integer n and every positive integer d, there exist UNIQUE integers q and r such that:

$$n = dq + r \quad \text{with} \quad 0 \leq r < d$$

Quotient q

The integer number of complete divisor-sized groups.

Remainder r

The leftover amount, always from 0 through d − 1.

Examples: the conditions matter

n	d	q	r	Verification
54	4	13	2	$54 = 4(13) + 2$
-54	4	-14	2	$-54 = 4(-14) + 2$
54	70	0	54	$54 = 70(0) + 54$
-17	5	-4	3	$-17 = 5(-4) + 3$

Check: $0 \leq r < d$ — even when n is negative.

Why is the quotient-remainder pair unique?

Suppose the same integer has two valid representations:

$$n = dq + r = dq' + r'$$

- Subtract: $d(q - q') = r' - r$.
- Both remainders lie between 0 and $d - 1$.
- Hence $|r' - r| < d$; the only multiple of d in that range is 0.
- Therefore $r = r'$ and $q = q'$.

Integer div and mod

$$n \operatorname{div} d = q \qquad n \bmod d = r$$

where $n = dq + r$, $d > 0$, and $0 \leq r < d$.

Example: 32 ÷ 9

$32 = 9(3) + 5$
$32 \operatorname{div} 9 = 3$

The remainder

$32 \bmod 9 = 5$
Remainder is nonnegative.

Mathematical mod versus programming %

Mathematics (Epp)

$-11 = 4(-3) + 1$
$-11 \text{ div } 4 = -3$
$-11 \bmod 4 = 1$

C++ / Java signed integers

$-11 / 4 = -2$
$-11 \,\%\, 4 = -3$
Integer division truncates toward zero.

The two conventions agree for nonnegative dividends and positive divisors.

Application: day of the week

Seven days form one complete cycle.

$$365 = 7(52) + 1$$

- After 364 days: the weekday is unchanged.
- After 365 days: the weekday moves forward by one.
- Only the remainder modulo 7 determines the shift.

Worked calendar example

On 16 October 2021 it was Saturday. What day was 20 February 2022?

$$15 + 30 + 31 + 31 + 20 = 127 \text{ days}$$

$$127 = 7(18) + 1$$

18 full weeks plus 1 day: Sunday.

Using a known remainder

Given m mod 11 = 6, find 4m mod 11.

$$m = 11q + 6$$

$$4m = 44q + 24 = 11(4q + 2) + 2$$

Because $0 \leq 2 < 11$, the remainder is 2.

Every integer is even or odd

$$n = 2q + r \qquad 0 \leq r < 2$$

Case 1: r = 0

n = 2q
Even integer

Case 2: r = 1

n = 2q + 1
Odd integer

Exactly one case applies to every integer.

Proving the parity of consecutive integers

Claim: For every integer m, m and m + 1 have opposite parity.

Case 1: m is even

$m = 2k$
$m + 1 = 2k + 1$ (odd)

Case 2: m is odd

$m = 2k + 1$
$m + 1 = 2(k + 1)$ (even)

Cases exhaust all integers, so the claim holds.

Proof method: divide into exhaustive cases

1 **State the claim and choose cases.**

2 Justify that all possibilities are covered.

3 Prove the claim separately in each case.

4 Combine the conclusions.

A case split is valid only when no possible input is omitted.

Four possible remainders modulo 4

$$n = 4q + r, \qquad r \in \{0, 1, 2, 3\}$$

r = 0

n = 4q + 0

r = 1

n = 4q + 1

r = 2

n = 4q + 2

r = 3

n = 4q + 3

The four cases are mutually exclusive and exhaustive.

A useful classification of odd integers

When divided by 4, an odd integer cannot leave remainder 0 or 2.

Odd case A

$n = 4q + 1$

Odd case B

$n = 4q + 3$

These two cases support proofs about odd squares.

STATEMENT

Theorem: squares of odd integers

Claim: Every odd integer has a square one more than a multiple of 8.

If n is odd, then $n^2 = 8m + 1$ for some integer m.

Proof plan: $n = 4q + 1$ OR $n = 4q + 3$.

Odd-square proof: first case

Case 1: $n = 4q + 1$

$$n^2 = (4q + 1)^2$$

$$= 16q^2 + 8q + 1$$

$$= 8(2q^2 + q) + 1$$

Odd-square proof: second case

Case 2: $n = 4q + 3$

$$n^2 = (4q + 3)^2$$

$$= 16q^2 + 24q + 9$$

$$= 8(2q^2 + 3q + 1) + 1$$

Concluding the proof

In either case, the term multiplying 8 is an integer:

Case 1

$m = 2q^2 + q$

Case 2

$m = 2q^2 + 3q + 1$

Therefore, $n^2 = 8m + 1$ for every odd integer n.

Absolute value: definition

For every real number x:

$$|x| = x \quad \text{if } x \geq 0; \qquad |x| = -x \quad \text{if } x < 0$$

Positive / zero

$|2| = 2,\ \ |0| = 0$

Negative

$|-2| = 2$

Absolute value is nonnegative

$$|x| \geq 0 \quad \text{and} \quad |x| = 0 \text{ iff } x = 0$$

If $x \geq 0$

$|x| = x \geq 0$.

If $x < 0$

$|x| = -x > 0$.

This follows directly from the piecewise definition.

Absolute-value bounds

$$-|r| \le r \le |r|$$

Case 1: $r \ge 0$

$|r| = r$. Therefore $-|r| \le r = |r|$.

Case 2: $r < 0$

$|r| = -r$. Therefore $-|r| = r < |r|$.

Do not replace $\le$ by $<$: equality occurs at a boundary.

Absolute value is unchanged by negation

$$|-r| = |r|$$

$r \geq 0$

$|r| = r$; $|-r| = r$.

$r < 0$

$|r| = -r$; $|-r| = -r$.

Both sides measure the same distance from zero.

The triangle inequality

$$|x + y| \leq |x| + |y|$$

Case 1: $x + y \geq 0$

$|x + y| = x + y \leq |x| + |y|$.

Case 2: $x + y < 0$

$|x + y| = -x - y \leq |x| + |y|$.

Example: $|-3 + 5| = 2 \leq 3 + 5 = 8$.

Concept check: find the invalid argument

Student claim

“$-19 = 6(-3) - 1$, so $-19 \bmod 6 = -1$.”

What condition has been violated?

$$0 \leq r < d$$

Core concepts and common pitfalls

- $n = dq + r$; d must be positive, and $0 \leq r < d$.
- The quotient and remainder are unique.
- Possible remainders modulo d are $0, 1, \ldots, d - 1$.
- Cases in a proof must cover every possibility.
- Absolute value is nonnegative; $\leq$ may be an equality.

Conceptual connections

Divisibility: $d \mid n \iff n \bmod d = 0$

Parity: $n \bmod 2$ classifies even / odd

Modulo 4: odd $n = 4q + 1$ OR $4q + 3$

Proof by cases: exhaustive remainders give exhaustive cases

Absolute value: sign-based case split proves inequalities

Worked examples: contrasting cases

Valid negative dividend

$-17 = 5(-4) + 3$; remainder 3.

Invalid proposed remainder

$-17 = 5(-3) - 2$; $r < 0$, invalid.

Zero remainder

$21 = 7(3) + 0$; 7 divides 21.

Boundary: $d > n \geq 0$

$5 = 9(0) + 5$; $q = 0$, $r = 5$.

Knowledge check — questions 1–4

- 1 Compute $-29 \operatorname{div} 6$ and $-29 \bmod 6$.

Knowledge check — questions 5–8

Answer key — questions 1–4

- 1 $-29 = 6(-5) + 1$. Quotient -5, remainder 1.
- 2 False. The upper bound is strict: $r < d$.
- 3 B. Option A has the wrong general sign; C excludes zero.
- 4 $m = 7q + 3 \Rightarrow 5m = 7(5q + 2) + 1$. Result: 1.

Answer key — questions 5–8

- 5 Split $m = 2k$ and $m = 2k + 1$; $m + 1$ flips parity.
- 6 Remainder -1 violates $0 \leq r < 4$. Correct: $-13 = 4(-4) + 3$.
- 7 Even $n = 2k \Rightarrow n^2$ divisible by 4; odd $n = 2k+1 \Rightarrow n^2 = 4(k^2+k)+1$.
- 8 False. $x = 1$, $y = -1$ gives $0 \neq 2$; the $\leq$ inequality holds.

Optional enrichment: why remainders matter

Circular indexing

To rotate through seven weekdays or a fixed-size circular buffer, reduce offsets modulo the cycle length.

Computing and security

Remainder classes appear in hash bucket indexing and modular computations used in cryptography.

Next: sequences and mathematical induction extend reasoning to infinitely many cases.

References and course resources

- Primary textbook: Susanna S. Epp, Discrete Mathematics with Applications, 4th ed.
- Original course materials: Prof. Mustafa Jarrar — www.jarrar.info/courses/DMath/
- Updated course resources: github.com/RuwaYafa/BZU_Courses/tree/main/COMP233
- Original slides circulated with assistance from Dr. Radi Jarrar.

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